

SIMILAR AEROBIC ADAPTATIONS WHEN TOTAL DAILY WORKLOAD IS THE SAME

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October 4, 2025

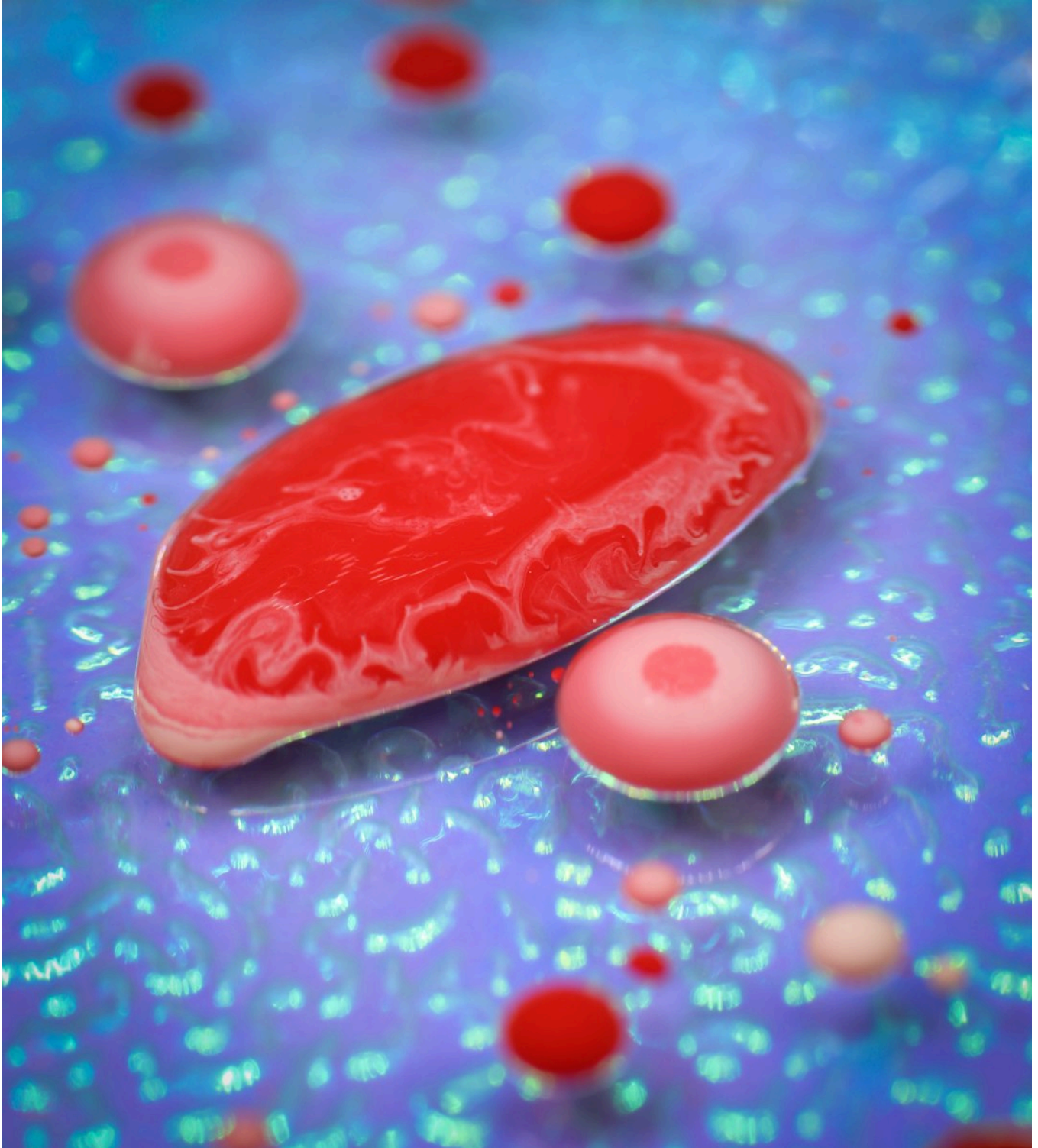


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In the last years, the **Norwegian Double-Threshold Days** have become very popular. This method consists of doing two sessions at medium/high intensity (the majority of Norwegians do them at an intensity between 2.0-4.0 mmol/L of blood lactate) within the same day: one in the morning and one in the afternoon. As a consequence of the great success Norwegian endurance athletes achieved in the last years, performing multiple daily training session has become quite attractive within the cycling world.

So, it can be interesting to see some scientific data which can help us to learn more about this training strategy.

When the total daily workload is the same, do multiple daily training sessions induce greater physiological adaptations compared to performing only one?

A study published by Mendes and Colleagues in 2022 on *Scientific Reports* (Federal University of the Jequitinhonha and Mucuri Valleys, Brazil) can help to answer this question. (1)

WHAT DID THEY DO?

30 Male Wistar Rats (!) were divided into 3 different groups, which for 8 weeks trained in three different ways:

1. **Control Group:** they did not perform any training.
2. **1x Day High Intensity Training (HIT) Group:** 5 times per week, they performed single daily sessions of 15 min treadmill run, which consisted of:
 - 3 min of warm-up, followed by six bouts of 1 min at 85-100% of speed at VO₂max interspersed by 1 min of passive recovery.
3. **3x Day High Intensity Training (HIT) Group:** 5 times per week, they performed three daily sessions of 5 min with an interval of 4 h between sessions (8AM, 12PM and 4 PM), which consisted of:
 - 1 min of warm-up, followed by two bouts of 1 min at 85-100% of speed at VO₂max interspersed by 1 min of passive recovery (total of 5 min).

As you can see, the overall daily duration, intensity and work done was similar between 1xDay HIT Group and 3xDay HIT Group.

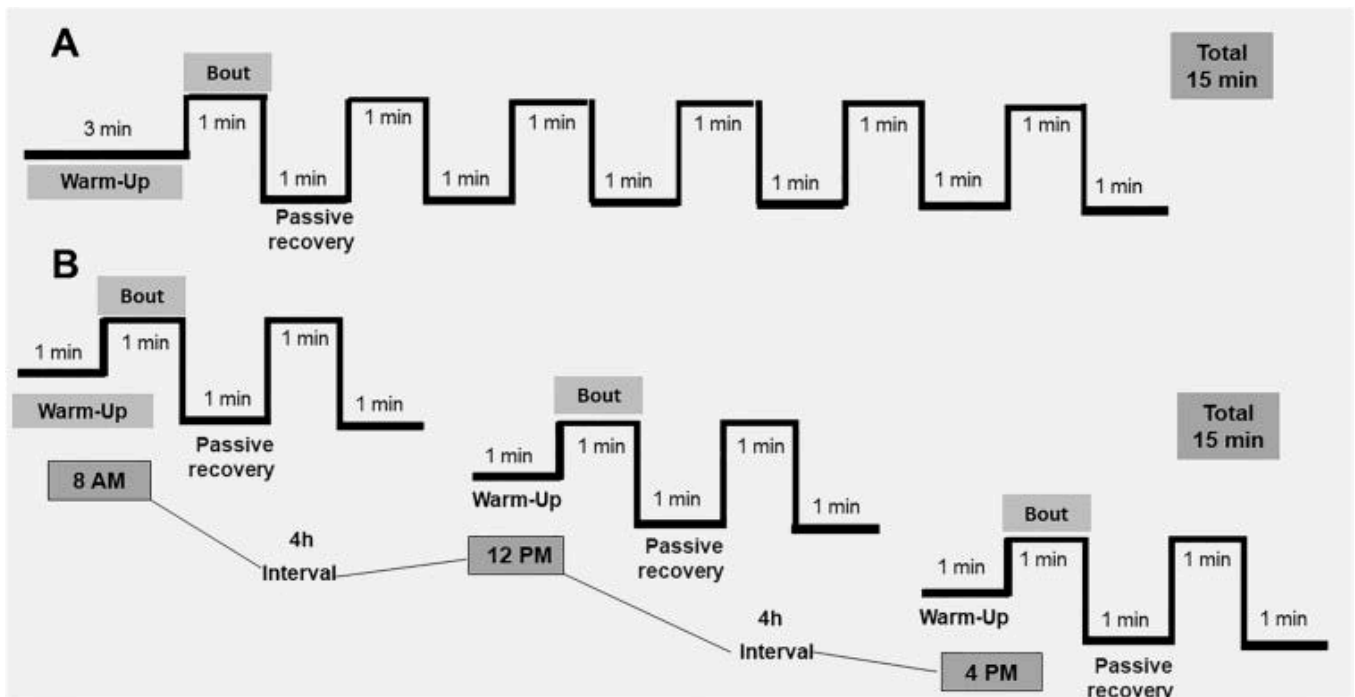


Image from Mendes et al. (1).

WHAT DID THEY FIND?

- The maximal oxygen consumption (VO_{2max}) resulted unchanged in the Control Group, while improved in a similar between the 1xDay and the 3x Day Group:
 - Control Group: from 52 ± 6 to 50 ± 3 ml/min/kg
 - 1xDay HIT Group: from 49 ± 4 to 68 ± 5 ml/min/kg
 - 3xDay HIT Group: from 52 ± 5 to 67 ± 3 ml/min/kg
- In line with VO_{2max} changes, the mitochondrial density (index of muscular aerobic capacity) of the gastrocnemius muscle improved similarly between the 1xDay and 3xDay HIT Groups.

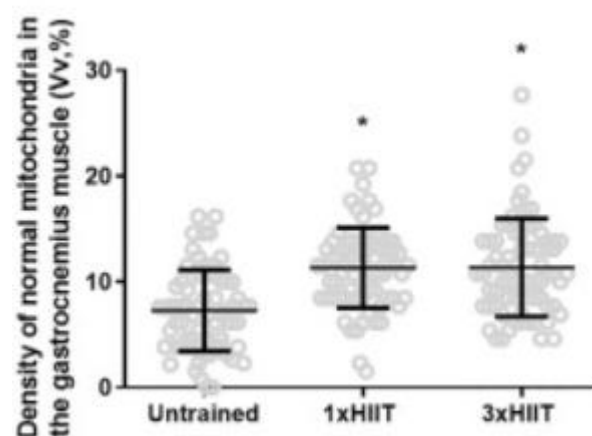


Image from Mendes et al. (1).

PRACTICAL TAKEAWAYS

When the total daily workload (both volume and intensity) is the same, multiple daily HIT training sessions induce similar aerobic adaptations when compared to performing only one.

GABRIELE'S COMMENT

No much more to add. Even if the study is on rats and not humans, and the workouts were very shorts, I think these data can be used as a 'proof of concept':

When the total daily workload (both volume and intensity) is the same, multiple daily HIT training sessions induce similar aerobic adaptations when compared to performing only one.

Please, note the premise: 'When total daily workload is the same'.

So, these data don't mean that double-threshold days are not useful and effective at all. Can they permit to accumulate more daily volume at high intensity (and so stimulate greater adaptations) without compromising recovery? If yes, in which scenario can they be more effective?

We will see this and more about double-threshold days in the next KIW Research Note!

Thanks for Reading! Share with your Team!

Gabriele Gallo, PhD in Exercise and Sport Science

REFERENCES:

1. Mendes BF, Costa-Pereira LV, de Andrade JA, Diniz Magalhães CO, de Pereira RRS, Esteves EA, Cassilhas RC, Andrade EF, Gripp F, de Magalhães FC, Honorato Sampaio K, Amorim FT, Dias-Peixoto MF. Superior cardiometabolic and cellular adaptive responses to multiple versus single daily sessions of high-intensity interval training in Wistar rats. Sci Rep. 2022 Dec 7;12(1):21187.



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